

THE SOLUTION TO KADISON'S PROBLEM ON ORTHONORMAL BASES OF UNITARIES FOR TYPE II_1 FACTORS

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1. STATEMENT OF THE THEOREM

Let (M, τ) be a finite von Neumann algebra with a faithful normal tracial state. We write

$$\mathcal{S}_0(M) = \{s \in M : s = s^* = s^{-1}, \tau(s) = 0\}$$

and let $L^2(M)_{\text{sa}}$ denote the real Hilbert space obtained by completing M_{sa} in the tracial L^2 -norm. For a nonzero projection e , let $\tau_e = \tau(e)^{-1}\tau|_{eMe}$.

Theorem 1.1. *Let (M, τ) be diffuse and put $\kappa = \text{dens } L^2(M, \tau)$. The following are equivalent.*

- (1) $L^2(M, \tau)$ has an orthonormal basis consisting of unitaries in M .
- (2) $L^2(M, \tau)$ has an orthonormal basis consisting of self-adjoint unitaries in M .
- (3) $L^2(M, \tau)$ has an orthonormal basis of the form

$$\{1\} \cup B, \quad B \subset \mathcal{S}_0(M).$$

- (4) For every nonzero central projection $z \in Z(M)$,

$$\text{dens } L^2(zM, \tau_z) = \kappa. \tag{CD}$$

In particular, every type II_1 factor has an orthonormal basis of unitaries.

The separable case of (4) $\Rightarrow$ (3) is Theorem 1 of [HTZ26]. We therefore prove only the case $\kappa > \aleph_0$.

2. ELEMENTARY FACTS

We use the trace-preserving conditional expectation $E_N : M \rightarrow N$ for a von Neumann subalgebra $N \subset M$; its L^2 -extension is the orthogonal projection onto $L^2(N)$ [Tak72]. We also use Kaplansky's density theorem in the tracial standard representation [KR97a, p. 329, Theorem 5.3.5].

Lemma 2.1. *Let M be finite tracial.*

- (1) For $F \subset M$,

$$\text{dens } L^2(W^*(F)) \leq \max\{|F|, \aleph_0\}.$$

- (2) If $(M_i)_{i \in I}$ is an increasing directed family and $Q = W^*(\bigcup_i M_i)$, then

$$L^2(Q) = \overline{\bigcup_i L^2(M_i)}^{\|\cdot\|_2}.$$

- (3) Every von Neumann subalgebra $N \subset M$ has a self-adjoint generating set of cardinality at most $\text{dens } L^2(N)$.
- (4) If T is an orthonormal family in a Hilbert space, then

$$\text{supp}_T(\xi) := \{t \in T : \langle \xi, t \rangle \neq 0\}$$

is countable for every vector ξ .

Moreover,

$$M \cap L^2(N) = N. \tag{2.1}$$

Proof. Parts (i)–(iii) follow from rational $*$ -polynomial approximation and Kaplansky density; (iv) is Bessel's inequality. For (2.1), if $x \in M \cap L^2(N)$, then $x = E_N(x) \in N$. $\square$

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Lemma 2.2. *In the tracial standard representation, the trace vectors are precisely the vectors represented by unitaries in M .*

Proof. This is the standard trace-vector identification: if ξ is a trace vector, the map $x \mapsto x\xi$ extends to an isometry commuting with the left action; the commutant is the algebra of right multiplications, which is finite, hence this isometry is unitary. See [KR97b]. $\square$

Lemma 2.3. *Let $e \in M$ be nonzero and let $c(e)$ be its central support. Then*

$$\text{dens } L^2(eMe, \tau_e) = \text{dens } L^2(c(e)M, \tau_{c(e)}).$$

Consequently, if (CD) holds and $\kappa > \aleph_0$, then every nonzero corner eMe has L^2 -density κ .

Proof. Put $z = c(e)$ and $\mu = \text{dens } L^2(eMe)$. The L^2 -closed linear span of MeM is $zL^2(M)$: the orthogonal projection onto this closed two-sided invariant subspace is multiplication by a central projection, and the defining property of $c(e)$ forces that projection to be z .

Choose a sequence of finite sums from MeM converging to z in L^2 , and enumerate all coefficients occurring in that sequence as two countable families (a_i) and (b_j) . Let K be the closed linear span of

$$\{a_i x b_j : i, j \in \mathbb{N}, x \in eMe\}.$$

We claim that $K = zL^2(M)$. Certainly $K \subset zL^2(M)$. Fix $x \in zM$. Multiplying the chosen approximation of z on the left by a term $a_i e b_j x$ shows that $a_i e b_j x \in K$, because every resulting product has the form

$$a_i e b_j x a_j e b_j = a_i (e b_j x a_j e) b_j.$$

Multiplying the same approximation of z on the right by x now gives $x = zx \in K$. Thus $K = zL^2(M)$. Each map $x \mapsto a_i x b_j$ is L^2 -continuous. Hence the images, under this countable family of maps, of an L^2 -dense subset of eMe of cardinality μ have rational complex span dense in $zL^2(M)$. Thus

$$\text{dens } L^2(zM) \leq \mu.$$

The reverse inequality follows from the corner inclusion. Normalizing the traces does not change density characters. $\square$

Lemma 2.4. *Let $\mathcal{V} \subset \mathcal{U}(M)$ be orthonormal. For every nonzero central projection z ,*

$$|\mathcal{V}| \leq \text{dens } L^2(zM, \tau_z).$$

Proof. Let D be dense in $zL^2(M)$ with $|D| = \mu$. Since $\|zu\|_2^2 = \tau(z) > 0$ for every $u \in \mathcal{V}$, some $d \in D$ satisfies $\langle d, u \rangle \neq 0$. Hence

$$\mathcal{V} = \bigcup_{d \in D} \text{supp}_{\mathcal{V}}(d).$$

Each support is countable by Lemma 2.1(iv), so $|\mathcal{V}| \leq \mu$. $\square$

3. PROJECTION LIFTING AND FINITE APPROXIMATION

For the remainder of Sections 3–5, assume (CD) and $\kappa = \text{dens } L^2(M) > \aleph_0$.

Lemma 3.1 (Projection lifting). *Let $N \subset M$ be a von Neumann subalgebra such that*

$$\text{dens } L^2(eMe) > \text{dens } L^2(N) \quad (0 \neq e \in \mathcal{P}(M)).$$

For every positive contraction $a \in N$, there is a projection $p \in M$ with

$$E_N(p) = a.$$

Proof. Set

$$K = \{x \in M : 0 \leq x \leq 1, E_N(x) = a\}.$$

It is nonempty, convex, and ultraweakly compact. Let p be an extreme point. Suppose that p is not a projection. Then for some $\varepsilon \in (0, 1/2)$ the spectral projection

$$e = 1_{[\varepsilon, 1-\varepsilon]}(p)$$

is nonzero. Put $D = W^*(N \cup \{e\})$. Lemma 2.1 gives

$$\text{dens } L^2(D) \leq \text{dens } L^2(N),$$

so $eMe \not\subset D$. Choose $b = b^* \in eMe \setminus D$ and set

$$h = b - E_D(b).$$

Then $h \neq 0$, $h = h^* \in eMe$, and $E_N(h) = 0$. After rescaling, $\|h\|_\infty < \varepsilon$. Since e commutes with p and

$$\varepsilon e \leq epe \leq (1 - \varepsilon)e,$$

both $p + h$ and $p - h$ lie in K , contradicting extremality. Thus p is a projection. This is the density-gap version of the standard extreme-point argument of [DFHS12, Lemma 2.1 and Remark 2.3(iii)]. $\square$

Lemma 3.2 (Norming symmetry). *Let $N \subset M$ satisfy $\text{dens } L^2(N) < \kappa$, let $\Gamma \subset M_{\text{sa}}$ have cardinality $< \kappa$, and let $0 \neq r = r^* \in M$ satisfy*

$$E_N(r) = 0, \quad r \perp \Gamma.$$

Then there is $s \in \mathcal{S}_0(M)$ such that

$$E_N(s) = 0, \quad s \perp \Gamma, \quad \langle r, s \rangle_2 = \frac{\|r\|_2^2}{\|r\|_\infty}.$$

Proof. Let $D = W^*(N \cup \Gamma \cup \{r\})$. By Lemma 2.1, $\text{dens } L^2(D) < \kappa$. Lemma 2.3 therefore implies that every nonzero corner has strictly larger density than $L^2(D)$. Apply Lemma 3.1 to

$$q = \frac{1}{2} \left(1 + \frac{r}{\|r\|_\infty} \right) \in D.$$

There is a projection p with $E_D(p) = q$. Put $s = 2p - 1$. Then

$$E_D(s) = \frac{r}{\|r\|_\infty}.$$

Applying E_N gives $E_N(s) = 0$; pairing with Γ gives $s \perp \Gamma$; pairing with r gives the displayed norming identity. Trace preservation gives $\tau(s) = 0$. $\square$

Lemma 3.3 (Finite approximation). *Let $N \subset M$ satisfy $\text{dens } L^2(N) < \kappa$. Let $\Gamma \subset \mathcal{S}_0(M)$ satisfy $|\Gamma| < \kappa$ and $E_N(\Gamma) = 0$. Let V be the real span of finitely many orthonormal symmetries orthogonal to $L^2(N)$ and to Γ .*

If $b = b^ \in M$ satisfies $E_N(b) = 0$ and $b \perp \Gamma$, then for every $\varepsilon > 0$ there is a finite orthonormal family $F \subset \mathcal{S}_0(M)$, orthogonal to $L^2(N)$, Γ , and V , such that*

$$\text{dist}(b, V + \text{span}_{\mathbb{R}} F) < \varepsilon.$$

Proof. Let $r_0 = b - P_V b$. If $\|r_0\|_2 < \varepsilon$, take $F = \emptyset$. Otherwise, while $\|r_{j-1}\|_2 \geq \varepsilon$, apply Lemma 3.2 with the previously chosen symmetries added to the constraints. We obtain a new symmetry u_j and define

$$a_j = \frac{\|r_{j-1}\|_2^2}{\|r_{j-1}\|_\infty}, \quad r_j = r_{j-1} - a_j u_j.$$

The vectors u_j are mutually orthogonal and satisfy all required orthogonality conditions. Since $\langle r_{j-1}, u_j \rangle = a_j$,

$$\|r_j\|_2^2 = \|r_{j-1}\|_2^2 - a_j^2, \quad \sum_{i=1}^j a_i^2 \leq \|r_0\|_2^2.$$

Moreover,

$$\|r_{j-1}\|_\infty \leq \|r_0\|_\infty + \sum_{i < j} |a_i| \leq \|r_0\|_\infty + \sqrt{j-1} \|r_0\|_2.$$

If the construction never stopped, then with $C = \|r_0\|_\infty + \|r_0\|_2$ we would have

$$a_j^2 \geq \frac{\varepsilon^4}{C^2 j},$$

contradicting $\sum_j a_j^2 < \infty$. Hence it stops after finitely many steps, and the resulting family F has the required property. $\square$

4. CLOSING COEFFICIENT SUPPORTS

Let $\Omega \subset \mathcal{S}_0(M)$ be orthonormal and write P_Ω for the real orthogonal projection onto $\overline{\text{span}}_{\mathbb{R}}\Omega$. We say that Ω has the bounded-projection property if

$$P_\Omega(M_{\text{sa}}) \subset M_{\text{sa}}. \quad (\text{BP})$$

For $F \subset M_{\text{sa}}$, let $\mathcal{W}(F)$ be the real and imaginary parts of all unital $*$ -polynomials over $\mathbb{Q} + i\mathbb{Q}$ in finitely many elements of F .

Proposition 4.1 (Support closure). *Assume that Ω satisfies (BP). There is a closure operation cl_Ω on subsets of Ω such that*

(1)

$$\text{cl}_\Omega(\emptyset) = \emptyset, \quad |\text{cl}_\Omega(S)| \leq \max\{|S|, \aleph_0\};$$

(2) *for every nonempty directed increasing family (S_i) ,*

$$\text{cl}_\Omega\left(\bigcup_i S_i\right) = \bigcup_i \text{cl}_\Omega(S_i);$$

(3) *every closed S also satisfies (BP).*

If $\Omega \subset \Omega'$ and both satisfy (BP), a closure on Ω' can be chosen compatibly with any prescribed closure on Ω , in the sense that

$$S \text{ closed in } \Omega' \implies S \cap \Omega \text{ closed in } \Omega.$$

Proof. For $S \subset \Omega$, let $S_0 = S$ and

$$S_{n+1} = S_n \cup \bigcup_{w \in \mathcal{W}(S_n)} \text{supp}_\Omega(P_\Omega w), \quad c_\Omega(S) = \bigcup_{n \geq 0} S_n.$$

Every support is countable and every polynomial has finitely many inputs. Hence

$$|c_\Omega(S)| \leq \max\{|S|, \aleph_0\}.$$

The same finite-input observation gives monotonicity, idempotence, and preservation of directed unions. Since every element of Ω has trace zero, scalar polynomials have zero Ω -projection, so $c_\Omega(\emptyset) = \emptyset$.

Suppose $S = c_\Omega(S)$ and put $R = W^*(S)$. By construction and Kaplansky density,

$$P_\Omega(L^2(R)_{\text{sa}}) \subset \overline{\text{span}}_{\mathbb{R}} S.$$

For $x \in L^2(M)_{\text{sa}}$ and $s \in S$,

$$\langle P_\Omega E_R x, s \rangle = \langle E_R x, s \rangle = \langle x, s \rangle.$$

The left-hand side lies in the closed span of S , and therefore

$$P_S = P_\Omega E_R. \quad (4.1)$$

For $x \in M_{\text{sa}}$, both $E_R x$ and $P_\Omega E_R x$ are bounded self-adjoint elements. Thus S satisfies (BP). We may therefore take $\text{cl}_\Omega = c_\Omega$.

For compatibility, let $c_{\Omega'}$ be the closure just constructed on Ω' and let cl_Ω be the prescribed old closure. Define

$$F(S) = c_{\Omega'}(S) \cup \text{cl}_\Omega(S \cap \Omega), \quad \text{cl}_{\Omega'}(S) = \bigcup_{n \geq 0} F^n(S).$$

The map F is extensive, monotone, preserves directed unions, fixes the empty set, and has the same cardinal bound. Hence its countable iteration is a closure with (i)–(ii). If S is closed, then $F(S) = S$, so S is $c_{\Omega'}$ -closed and $S \cap \Omega$ is cl_Ω -closed. The preceding argument gives (BP) for S . $\square$

5. RELATIVE EXTENSION

Proposition 5.1 (Small relative extension). *Let $N \subset M$ satisfy $\text{dens } L^2(N) < \kappa$. Let $\Omega \subset \mathcal{S}_0(M)$ be orthonormal, satisfy (BP), and satisfy*

$$|\Omega| < \kappa, \quad E_N(\Omega) = 0.$$

Fix a closure cl_Ω as in Proposition 4.1. Let $\aleph_0 \leq \lambda < \kappa$. If

$$S \subset \Omega, \quad X \subset M_{\text{sa}}, \quad Y \subset N_{\text{sa}}, \quad |S|, |X|, |Y| \leq \lambda,$$

then there are von Neumann algebras $C \subset N$, $C \subset R \subset M$, a closed set $\Theta \subset \Omega$, and an orthonormal family $U \subset \mathcal{S}_0(M)$ such that

$$S \subset \Theta, \quad X \subset R, \quad Y \subset C, \quad E_N(U) = 0, \quad U \perp \Omega, \\ |\Theta|, |U|, \text{dens } L^2(R) \leq \lambda,$$

and

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}}(\Theta \cup U)}. \quad (5.1)$$

Proof. We argue by induction on the infinite cardinal λ .

The countable case. Construct increasing countable sets $D_n \subset N_{\text{sa}}$, closed sets $\Theta_n \subset \Omega$, and finite orthonormal families $U_n \subset \mathcal{S}_0(M)$ with $E_N(U_n) = 0$ and $U_n \perp \Omega$. Start with

$$D_0 = Y \cup \{1\}, \quad \Theta_0 = \text{cl}_\Omega(S), \quad U_0 = \emptyset.$$

At stage n , put

$$G_n = D_n \cup X \cup \Theta_n \cup U_n$$

and enumerate the countable set $\mathcal{W}(G_n) \times \mathbb{N}$ as $(t_{n,j})_{j \in \mathbb{N}_0}$. Fix a bijection $\pi : \mathbb{N}_0^2 \rightarrow \mathbb{N}_0$ such that $\pi(m, j) \geq m$. At stage $n = \pi(m, j)$, process the task $t_{m,j} = (y, k)$. Thus every task created at any finite stage is eventually processed, and for every fixed polynomial y that occurs in some G_m , the tasks (y, k) are processed for all k .

When (y, k) is processed, set

$$b(y) = y - E_N(y) - P_\Omega y.$$

By (BP), $b(y)$ is bounded and self-adjoint. Because $E_N(\Omega) = 0$,

$$b(y) \perp L^2(N), \quad b(y) \perp \Omega.$$

Adjoin $E_N(y)$ to D_n , enlarge Θ_n by closing the countable set $\text{supp}_\Omega(P_\Omega y)$, and apply Lemma 3.3 to $b(y)$ with

$$\Gamma = \Omega, \quad V = \text{span}_{\mathbb{R}} U_n, \quad \varepsilon = 2^{-k}.$$

Adjoin the resulting finite family to U_n .

Let

$$D = \bigcup_n D_n, \quad \Theta = \bigcup_n \Theta_n, \quad U = \bigcup_n U_n, \\ C = W^*(D), \quad R = W^*(D \cup X \cup \Theta \cup U).$$

Directed continuity of the closure makes Θ closed. If

$$y \in \mathcal{W}(D \cup X \cup \Theta \cup U),$$

then all inputs of y occur at some finite stage, and the schedule processes (y, k) for every k . Hence

$$E_N(y) \in C, \quad P_\Omega y \in \overline{\text{span}_{\mathbb{R}} \Theta}, \quad b(y) \in \overline{\text{span}_{\mathbb{R}} U}.$$

These three subspaces are mutually orthogonal. Since such polynomial parts are L^2 -dense in $L^2(R)_{\text{sa}}$, (5.1) follows.

The induction step. Let $\lambda > \aleph_0$ and assume the proposition for every smaller infinite cardinal. List the tagged elements of S , X , and Y along λ , allowing repetitions. For $\alpha \leq \lambda$, put

$$\lambda_\alpha = \max\{\aleph_0, |\alpha|\}.$$

We construct increasing $C_\alpha \subset N$, $C_\alpha \subset R_\alpha \subset M$, closed $\Theta_\alpha \subset \Omega$, and orthonormal U_α such that

$$E_N(U_\alpha) = 0, \quad U_\alpha \perp \Omega, \\ L^2(R_\alpha)_{\text{sa}} = L^2(C_\alpha)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}}(\Theta_\alpha \cup U_\alpha)}, \quad (5.2)$$

all earlier tagged data are absorbed, and

$$|\Theta_\alpha|, |U_\alpha|, \text{dens } L^2(R_\alpha) \leq \lambda_\alpha.$$

Start with $C_0 = R_0 = \mathbb{C}1$ and $\Theta_0 = U_0 = \emptyset$.

Suppose the objects have been constructed at stage $\alpha < \lambda$. From (5.2),

$$P_{\Omega \cup U_\alpha} = P_\Omega + E_{R_\alpha} - E_{C_\alpha} - P_{\Theta_\alpha}.$$

Thus $\Omega \cup U_\alpha$ is an orthonormal family satisfying (BP). Moreover $|\Omega \cup U_\alpha| < \kappa$ and $E_N(\Omega \cup U_\alpha) = 0$. Choose on it a closure compatible with the fixed closure on Ω , using Proposition 4.1.

Choose bounded self-adjoint L^2 -dense sets in C_α and R_α of cardinality at most λ_α . Apply the induction hypothesis at the smaller cardinal $\lambda_\alpha < \lambda$ to the enlarged background $\Omega \cup U_\alpha$, taking as input the set $\Theta_\alpha \cup U_\alpha$, those two dense sets, and the next tagged datum. We obtain

$$C^+ \subset N, \quad C^+ \subset R^+ \subset M,$$

a closed $T \subset \Omega \cup U_\alpha$ containing $\Theta_\alpha \cup U_\alpha$, and an orthonormal family $V \perp \Omega \cup U_\alpha$ with $E_N(V) = 0$ such that

$$L^2(R^+)_{\text{sa}} = L^2(C^+)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}}}(T \cup V).$$

The dense input sets imply, using (2.1),

$$C_\alpha \subset C^+, \quad R_\alpha \subset R^+.$$

Set

$$C_{\alpha+1} = C^+, \quad R_{\alpha+1} = R^+, \quad \Theta_{\alpha+1} = T \cap \Omega, \quad U_{\alpha+1} = U_\alpha \cup V.$$

Compatibility makes $\Theta_{\alpha+1}$ closed. Since T contains U_α and is contained in $\Omega \cup U_\alpha$,

$$T = (T \cap \Omega) \cup U_\alpha,$$

so the displayed decomposition is exactly (5.2) at $\alpha + 1$.

At a nonzero limit $\delta \leq \lambda$, take generated unions for C_δ, R_δ and ordinary unions for Θ_δ, U_δ . Orthogonality and the identities $E_N(U_\alpha) = 0$ pass to the union. Lemma 2.1(ii) passes the decompositions to the L^2 -closures, and directed continuity makes Θ_δ closed. Since every previous stage is bounded by

$$\lambda_\delta = \max\{\aleph_0, |\delta|\},$$

the union of $|\delta|$ many dense sets, each of size at most λ_δ , still has size at most λ_δ . Thus all cardinal bounds persist. At $\delta = \lambda$ every input has been absorbed. This proves the proposition. $\square$

Proposition 5.2 (Local extension). *Let $N \subset M$, let $\aleph_0 \leq \lambda < \kappa$, and assume $\text{dens } L^2(N) \leq \lambda$. For every $X \subset M_{\text{sa}}$ with $|X| \leq \lambda$, there are*

$$N \cup X \subset P \subset M$$

and an orthonormal family $U \subset \mathcal{S}_0(P)$ such that

$$|U|, \text{dens } L^2(P) \leq \lambda, \quad L^2(P)_{\text{sa}} = L^2(N)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}}}U.$$

Proof. Choose a self-adjoint generating set Y for N with $|Y| \leq \lambda$. Apply Proposition 5.1 with $\Omega = S = \emptyset$ and inputs X, Y . The output satisfies $Y \subset C \subset N$, hence $C = N$. Set $P = R$. Finally, each $u \in U$ lies in $M \cap L^2(P) = P$ by (2.1). $\square$

Theorem 5.3 (Relative extension). *Assume (CD) and $\kappa > \aleph_0$. If $N \subset M$ satisfies*

$$\text{dens } L^2(N) < \kappa,$$

then there is an orthonormal family $U \subset \mathcal{S}_0(M)$ such that

$$L^2(M)_{\text{sa}} = L^2(N)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}}}U.$$

Proof. Choose a self-adjoint generating family $(g_\alpha)_{\alpha < \kappa}$ for M , allowing repetitions. Put

$$\mu_\alpha = \max\{\text{dens } L^2(N), \aleph_0, |\alpha|\}.$$

For every $\alpha < \kappa$, $\mu_\alpha < \kappa$. Starting with $N_0 = N$ and $U_0 = \emptyset$, recursively construct increasing von Neumann algebras N_α and orthonormal families $U_\alpha \subset \mathcal{S}_0(N_\alpha)$ such that

$$L^2(N_\alpha)_{\text{sa}} = L^2(N)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}}}U_\alpha, \quad \text{dens } L^2(N_\alpha) \leq \mu_\alpha,$$

and $g_\beta \in N_\alpha$ for every $\beta < \alpha$.

At a successor stage, apply Proposition 5.2 to N_α with $X = \{g_\alpha\}$ and $\lambda = \mu_\alpha$. The new complement supplied by that proposition is orthogonal to $L^2(N_\alpha)$, so adjoining it to U_α gives the required decomposition at $\alpha + 1$.

At a nonzero limit $\delta \leq \kappa$, set

$$N_\delta = W^*\left(\bigcup_{\alpha < \delta} N_\alpha\right), \quad U_\delta = \bigcup_{\alpha < \delta} U_\alpha.$$

Lemma 2.1(ii) gives the decomposition at the limit. The same common-bound argument as above yields

$$\text{dens } L^2(N_\delta) \leq \mu_\delta.$$

At the final stage, every generator belongs to N_κ , hence $N_\kappa = M$. Taking $U = U_\kappa$ proves the theorem. $\square$

6. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. The implications (3) $\Rightarrow$ (2) $\Rightarrow$ (1) are immediate.

Assume (1), and let $\mathcal{V}$ be an orthonormal basis of unitaries. Its cardinality is κ . For every nonzero central projection z , Lemma 2.4 gives

$$\kappa \leq \text{dens } L^2(zM, \tau_z).$$

The reverse inequality follows because multiplication by z is a bounded projection from $L^2(M)$ onto $zL^2(M)$. Hence (CD) holds.

Now assume (CD). If $\kappa = \aleph_0$, condition (3) is Theorem 1 and its proof in [HTZ26]. Suppose $\kappa > \aleph_0$. Apply Theorem 5.3 with $N = \mathbb{C}1$. Since

$$L^2(\mathbb{C}1)_{\text{sa}} = \mathbb{R}1,$$

there is an orthonormal family $U \subset \mathcal{S}_0(M)$ such that

$$L^2(M)_{\text{sa}} = \mathbb{R}1 \oplus \overline{\text{span}_{\mathbb{R}} U}.$$

Complexifying gives an orthonormal basis

$$\{1\} \cup U$$

of $L^2(M)$, proving (3).

If M is a factor, its only nonzero central projection is 1, so (CD) is automatic. The last assertion follows. By Lemma 2.2, the same basis also gives a trace-vector orthonormal basis in the tracial standard representation. $\square$

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